

# Winning Space Race with Data Science

Predicting Falcon 9 first-stage landing success

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**\$62M**

advertised SpaceX Falcon 9  
launch cost

**\$165M+**

typical cost of other providers

**90**

Falcon 9 launch records used in  
modeling

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IBM Applied Data Science Capstone  
Final Presentation

GitHub repository: <https://github.com/adilsk007/IBM-Applied-Data-Science-Capstone>

# Outline

How the project moves from raw launch data to a landing-success prediction model

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- 01 Business problem and analytical objective
- 02 Data collection, wrangling and feature engineering
- 03 EDA with visualization and SQL
- 04 Interactive analytics with Folium and Plotly Dash
- 05 Predictive modeling and model comparison
- 06 Final conclusions and GitHub deliverables

# Executive Summary

The capstone uses an end-to-end data science workflow to estimate booster recovery success

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**Objective:** Predict whether the Falcon 9 first stage will land successfully after launch.

**Business value:** Successful booster recovery drives the cost advantage of reusable launch vehicles.

**Workflow:** API data collection, web scraping, cleaning, SQL/visual EDA, Folium maps, Dash dashboard and machine learning.

**Key dashboard insight:** KSC LC-39A has both the largest number of successful launches and the highest success rate in the dashboard dataset.

**Model result:** all four classifiers tied at 83.3% test accuracy; Decision Tree achieved the highest cross-validation score.

**KSC LC-39A**

largest successful launches and highest rate

**B5**

booster category with highest dashboard  
success rate

**3-4k kg**

payload range with strongest success rate signal

# Business Problem

A competitor needs a data-driven way to estimate launch outcome and cost competitiveness

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## Scenario

SpaceY wants to compete with SpaceX. Since SpaceX can reuse first-stage boosters, predicting landing success helps estimate launch cost and bidding competitiveness.

## Analytical question

Can launch features such as payload mass, orbit, launch site, booster version and launch history predict whether the Falcon 9 first stage lands successfully?

## Target variable

**Class = 1**

Successful landing

**Class = 0**

Unsuccessful landing

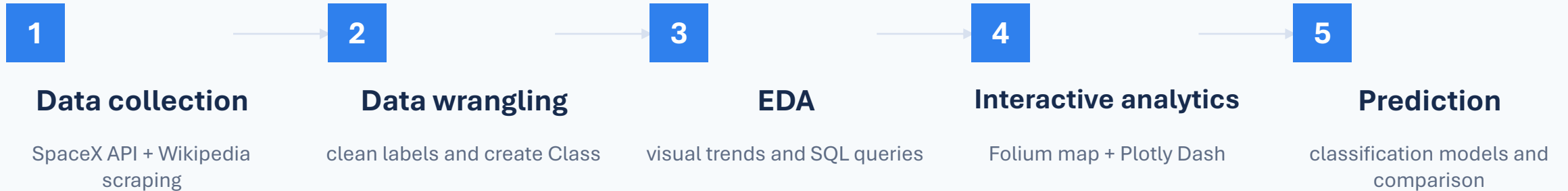
**Reusable**

first stage recovery changes  
launch economics

# Data Science Methodology

Each lab contributes one required part of the final capstone presentation

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- Each notebook was completed and saved as a solved version.
- Dashboard screenshots were captured for use in the final stakeholder presentation.
- Final deliverables: GitHub repository URL and PDF/PPT presentation.

# Data Collection

Launch records were collected from two complementary sources

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**SpaceX REST API:** launch, rocket, payload, launchpad and landing metadata.

**Wikipedia web scraping:** Falcon 9 / Falcon Heavy historical launch table.

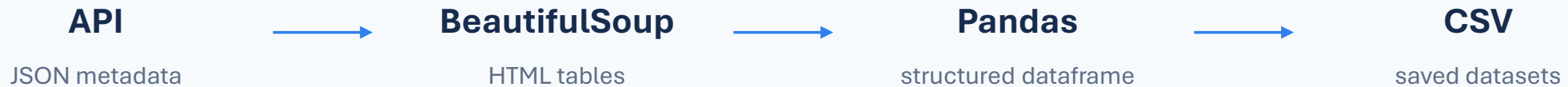
Data was converted into Pandas DataFrames and exported for later labs.

**dataset\_part\_1.csv**

cleaned API-derived launch dataset

**spacex\_web\_scraped.  
csv**

Wikipedia launch data export



# Data Wrangling

Landing outcomes were converted into a machine-learning target label

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- Filtered the launch data to Falcon 9 records only.
- Replaced missing PayloadMass values with the mean payload mass.
- Converted categorical landing outcomes into binary labels: 1 for successful landing and 0 for unsuccessful landing.
- Created dataset\_part\_2.csv for supervised learning and downstream analysis.

## Outcome mapping

True ASDS / True RTLS / True Ocean

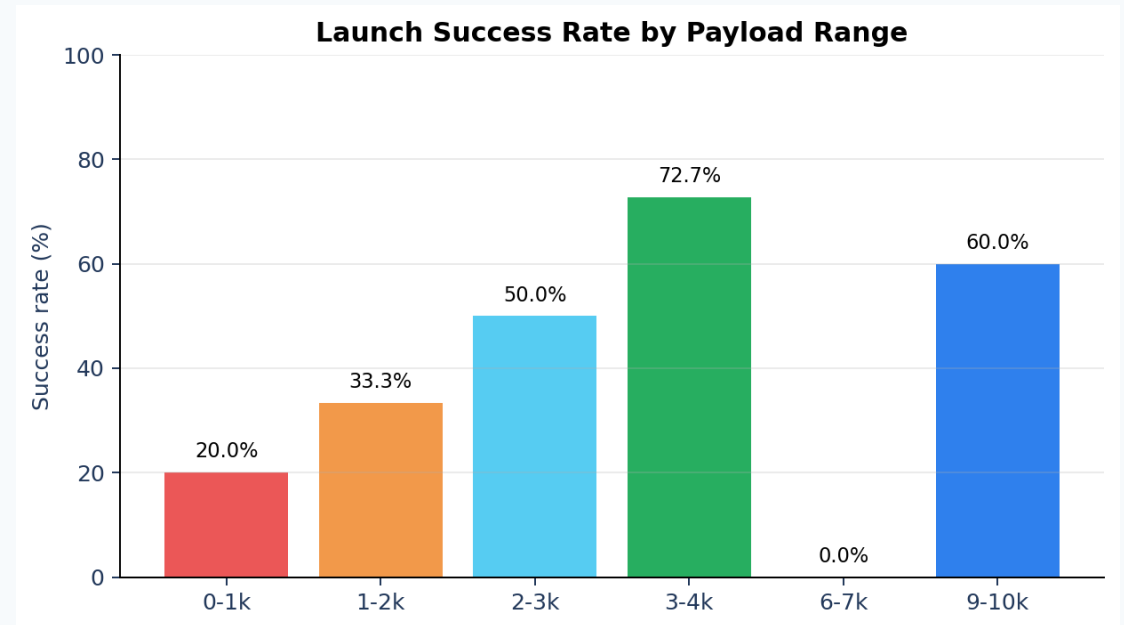
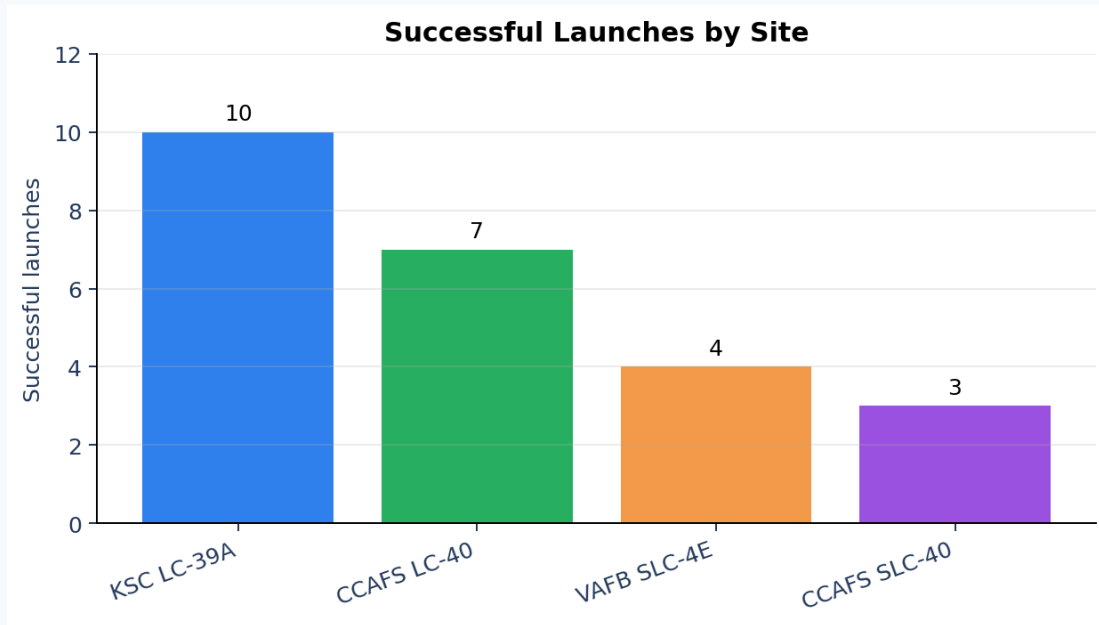
**Class = 1**

False ASDS / False RTLS / None / Ocean

**Class = 0**

# EDA with Visualization

Launch success varies by site, payload range and booster category



Insight: KSC LC-39A leads in successful launches, while payloads around 3,000-4,000 kg show the strongest success-rate signal in the dashboard data.

# EDA with SQL

SQL queries were used to validate launch-site, payload and outcome patterns

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## SQL question

## Result used in final analysis

**Unique launch sites**

4 launch sites

**NASA (CRS) payload mass**

45,596 kg total payload mass

**Average payload for F9 v1.1**

2,928.4 kg

**First successful ground-pad landing**

2015-12-22

**Mission outcome counts**

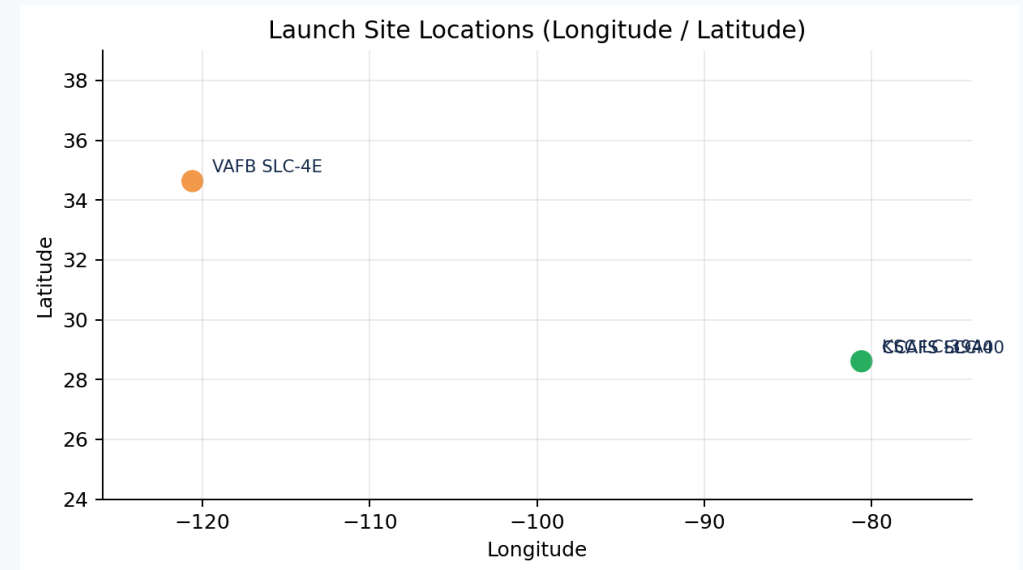
Mostly successful missions in the SQL dataset

**Why this matters:** SQL gives auditable, reproducible answers for the presentation and supports the visual analysis results.

# Interactive Map with Folium

Launch locations and nearby infrastructure were explored geographically

- All launch sites were plotted using latitude and longitude coordinates.
- Markers were colored by landing outcome: successful versus unsuccessful.
- Distance calculations were added from a launch site to the coastline and nearby infrastructure.
- The exported HTML map should be uploaded to GitHub with the notebooks.



The Folium notebook also exports an interactive HTML map with markers, marker clusters and distance annotations.

# Plotly Dash Dashboard

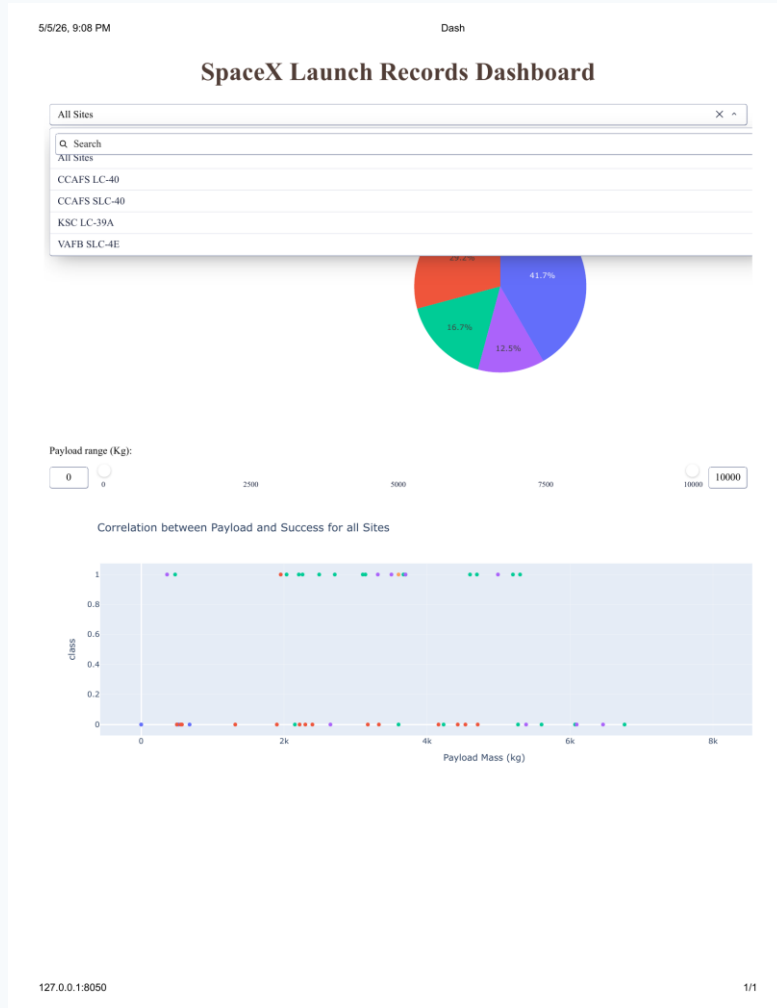
The dashboard enables interactive visual analytics in real time



- Dropdown lets the user select All Sites or an individual launch site.
- Pie chart updates to show total successful launches by site or success/failure split for one site.
- Payload slider filters the scatter plot by payload mass range.
- Scatter plot shows payload mass versus landing outcome, colored by booster version.

# Dashboard Interaction: Launch Site Dropdown

The launch-site selector changes both the pie chart and scatter plot context



- All Sites option shows the aggregate view across the dataset.
- Specific launch-site selection supports a detailed success-rate comparison.
- This screenshot proves Task 1 and Task 2 dashboard requirements are implemented.

4

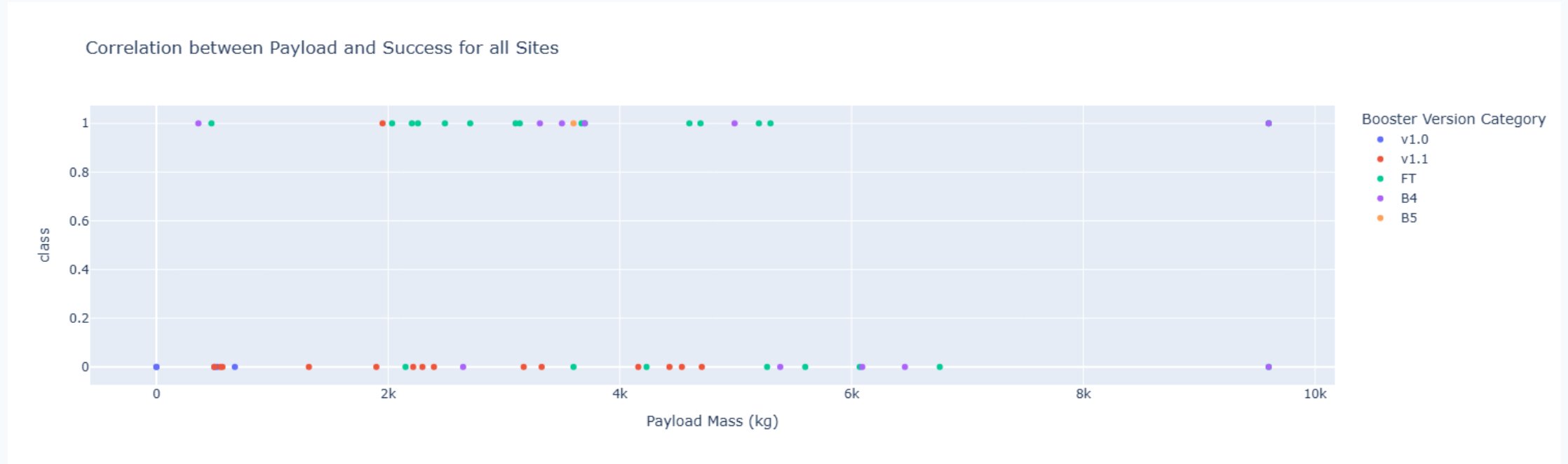
available launch-site filters

ALL

default dashboard selection

# Payload Scatter Plot: All Sites

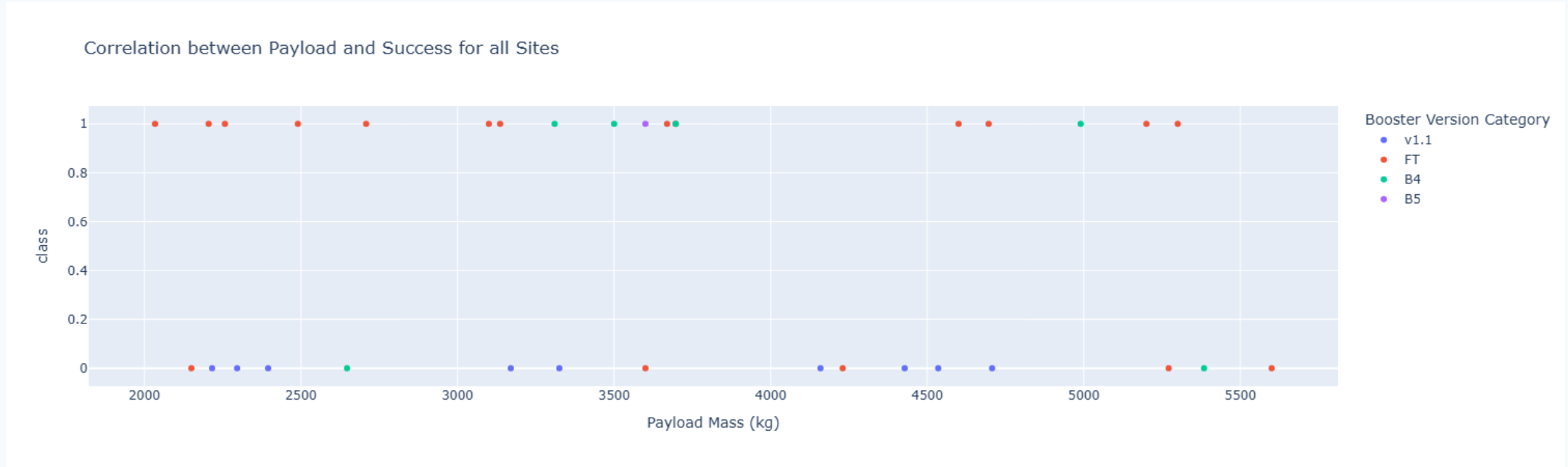
Each point represents one launch; class 1 means successful landing and class 0 means unsuccessful landing



Interpretation: because the target is binary, points naturally form two horizontal bands. The color label shows booster version category.

# Payload Slider Filter

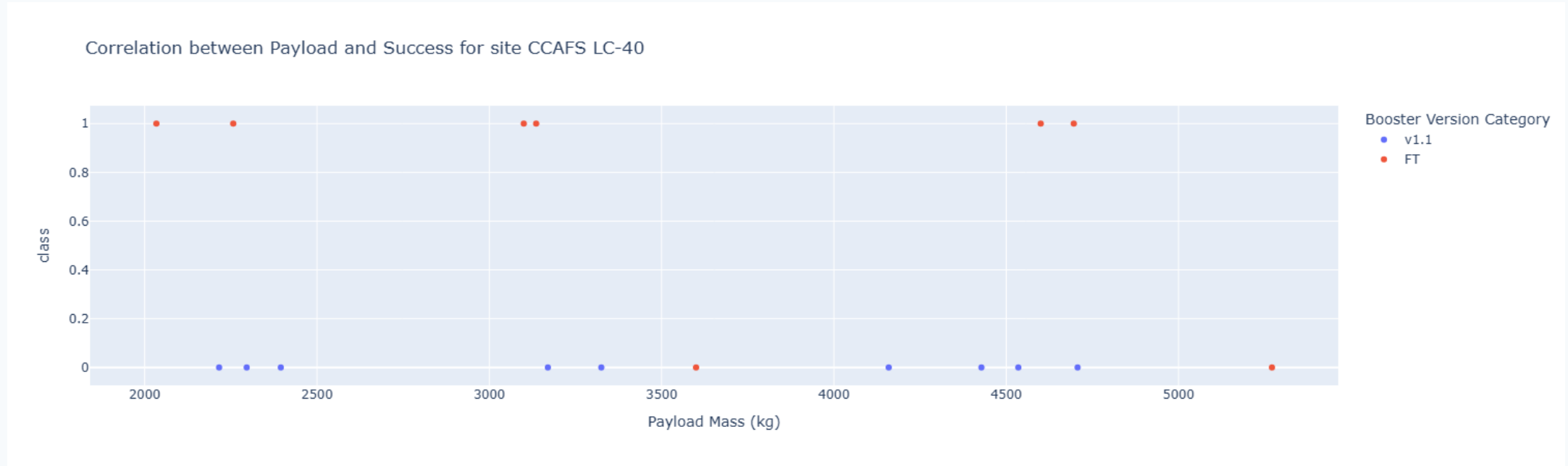
Changing the payload range filters the scatter plot without changing the app code



Filtered view focuses on payloads in the mid-range band, approximately 2,000-6,000 kg. The dashboard can be used to visually compare high-success and low-success payload ranges. This screenshot demonstrates the payload slider and scatter callback working together.

# Selected Site View: CCAFS LC-40

Filtering by site provides a site-specific payload-to-outcome view

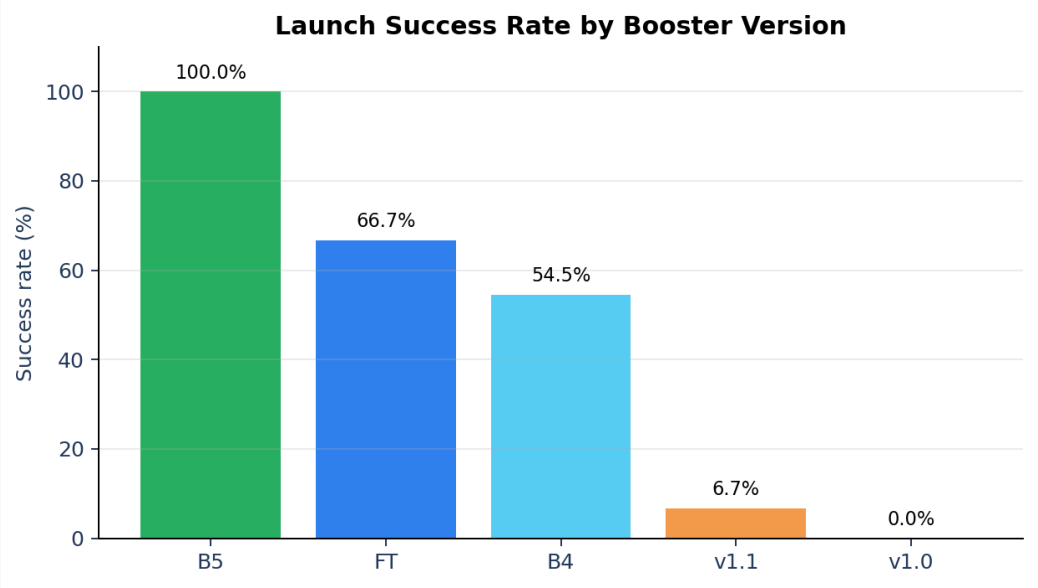


CCAFS LC-40 view shows fewer points than the all-sites view because the data is filtered to one launch site. The same binary class interpretation applies: 1 = landed successfully, 0 = did not land successfully.

# Dashboard Insights

Final visual answers extracted from the interactive dashboard

|                               |                                     |
|-------------------------------|-------------------------------------|
| Largest successful launches   | KSC LC-39A - 10 successful launches |
| Highest launch success rate   | KSC LC-39A - 76.9%                  |
| Highest payload success range | 3,000-4,000 kg - 72.7%              |
| Lowest payload success range  | 6,000-7,000 kg - 0%                 |
| Highest booster success rate  | B5 - 100%                           |



Note: percentages are based on the dashboard dataset used in the Dash lab.

# Predictive Analysis: Feature Preparation

Categorical and numerical launch features were prepared for supervised learning

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- Target label Y was created from the Class column.
- Feature matrix X was loaded from dataset\_part\_3.csv after one-hot encoding.
- Features were standardized using StandardScaler.
- Data was split into training and testing sets with test\_size = 0.2 and random\_state = 2.

## Feature examples

PayloadMass  
Flights  
Block  
ReusedCount  
Orbit\_\*  
LaunchSite\_\*  
Serial\_\*  
GridFins / Reused / Legs

**80/20**

train-test split

**4**

classification algorithms  
compared

# Classification Models and Hyperparameter Tuning

GridSearchCV was used to select strong model configurations

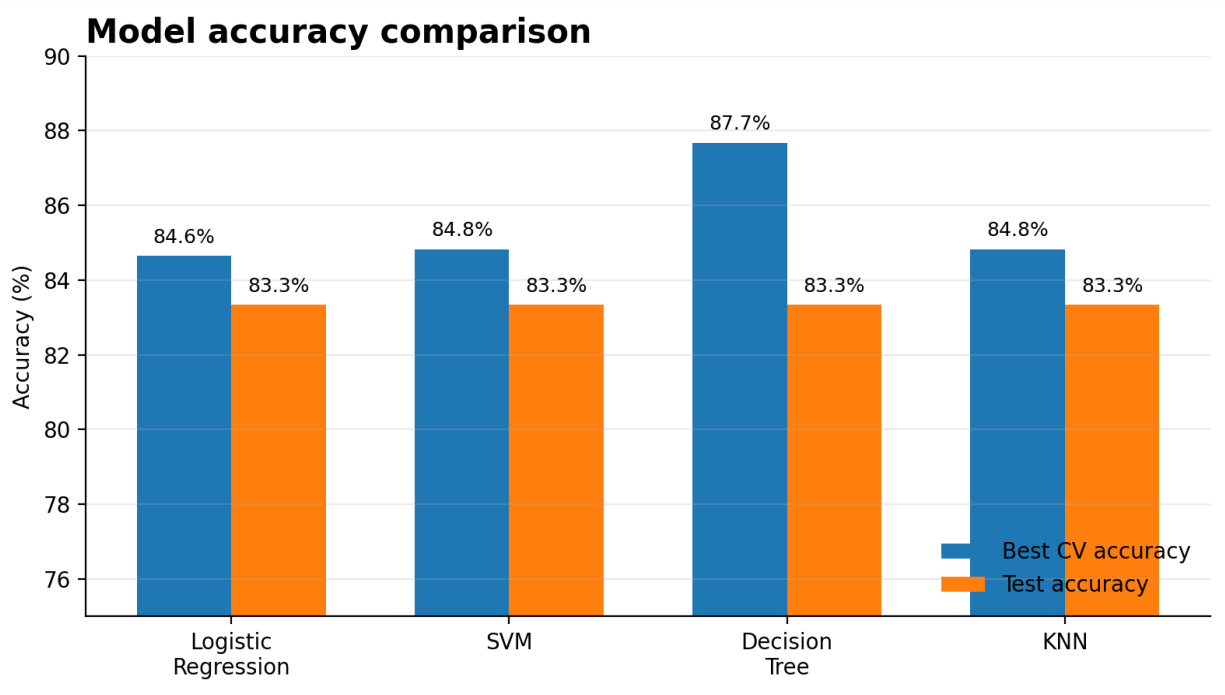
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| Model                  | Tuned parameters              | Purpose                     |
|------------------------|-------------------------------|-----------------------------|
| Logistic Regression    | C, penalty, solver            | linear baseline classifier  |
| Support Vector Machine | kernel, C, gamma              | nonlinear separation        |
| Decision Tree          | criterion, depth, split rules | interpretable rules         |
| K-Nearest Neighbors    | neighbors, distance metric    | similarity-based classifier |

- All models were evaluated on the same held-out test set.
- Models tied on test accuracy, so cross-validation score was also used to interpret model strength.
- All models were evaluated after hyperparameter tuning.

# Predictive Analysis Results

All models tied on test accuracy; Decision Tree achieved the highest cross-validation score



| Model               | Best CV | Test  |
|---------------------|---------|-------|
| Logistic Regression | 84.6%   | 83.3% |
| SVM                 | 84.8%   | 83.3% |
| Decision Tree       | 87.7%   | 83.3% |
| KNN                 | 84.8%   | 83.3% |

Test accuracy tie: all models - 83.3%

**Decision Tree** led cross-validation with 87.7%.

# Conclusion

Falcon 9 booster recovery can be predicted with useful accuracy from historical launch features

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- Launch site, payload mass range, booster version and launch history are important signals for landing success.
- KSC LC-39A stands out as the strongest dashboard site by successful launches and success rate.
- B5 booster category has the highest observed success rate in the dashboard analysis.
- Mid-range payloads around 3,000-4,000 kg show the strongest success-rate band in the dashboard results.
- All models reached 83.3% test accuracy; Decision Tree produced the strongest cross-validation score.

## Actionable

use the model to estimate recovery probability

## Reusable

landing success supports launch-cost estimation

## Portfolio-ready

upload notebooks, app and presentation to GitHub

# GitHub Repository Structure

Upload the solved notebooks, screenshots, datasets and final presentation

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```
IBM-Applied-Data-Science-Capstone/  
├── README.md  
├── final_presentation.pdf  
├── final_presentation.pptx  
├── notebooks/  
├── dashboard/  
├── data/  
└── images/
```

- README explains the business problem, workflow, tools and final deliverables.
- notebooks/ contains all solved .ipynb labs.
- dashboard/ contains the Dash notebook/app file.
- data/ contains dataset\_part\_1.csv, dataset\_part\_2.csv, dataset\_part\_3.csv and other generated CSV files.
- images/ contains the dashboard screenshots used in this presentation.

# Final Submission Checklist

Before submitting, make sure every required artifact is visible and easy to open

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- ✓ All solved notebooks run from top to bottom
- ✓ Dash dashboard screenshots saved as PNG
- ✓ Folium map exported as HTML or screenshot
- ✓ Final presentation exported to PDF
- ✓ GitHub repository is public
- ✓ Coursera submission includes GitHub URL and PDF

**Open your repository in a private/incognito browser window before submitting. If it opens without requiring login, the Coursera grader can access it.**

# Appendix: Dashboard Evidence

Screenshots captured from the implemented Dash application

```
jupyter spacex-dash-app-solved-notebook (autosaved)
```

```
File Edit View Insert Cell Kernel Widgets Help
```

```
Not Trusted JUPYTER
```

```
print(booster_success_rate.head(5))
```

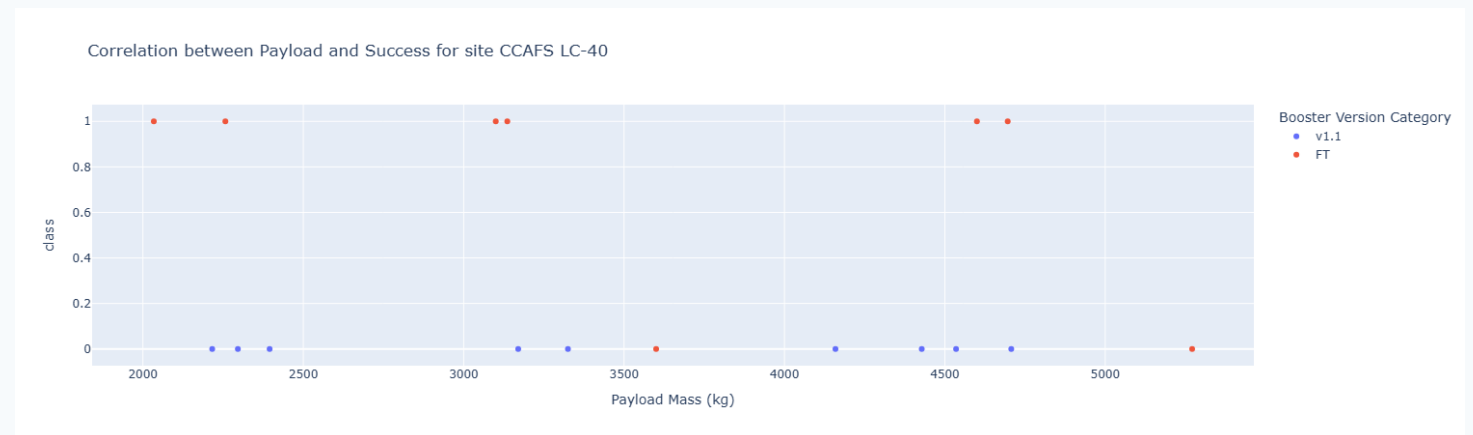
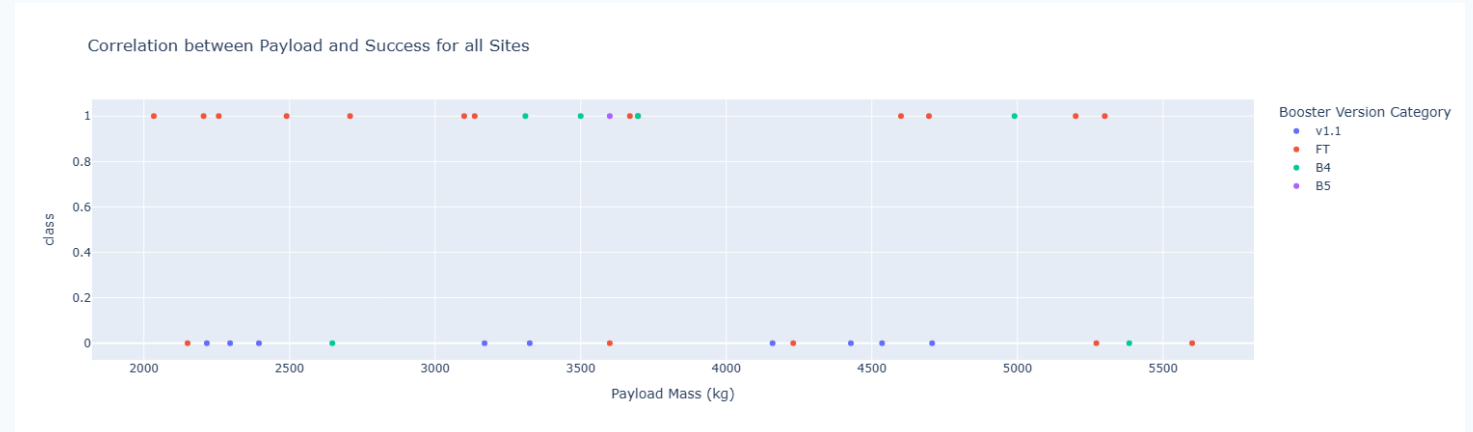
```
1. Site with largest successful launches:  
Launch Site  
KSC LC-39A 10  
dtype: int64
```

```
2. Site with highest launch success rate:  
Launch Site  
KSC LC-39A 0.769231  
Name: class, dtype: float64
```

```
3. Payload range(s) with highest success rate:  
Payload Range  
(3000.0, 4000.0] 0.727273  
(9000.0, 10000.0] 0.600000  
(2000.0, 3000.0] 0.500000  
Name: class, dtype: float64
```

```
4. Payload range(s) with lowest success rate:  
Payload Range  
(1000.0, 2000.0] 0.333333  
(-0.001, 1000.0] 0.200000  
(6000.0, 7000.0] 0.000000  
Name: class, dtype: float64
```

```
5. Booster version category with highest launch success rate:  
Booster Version Category  
B5 1.000000  
FT 0.666667  
B4 0.545455  
v1.1 0.066667  
v1.0 0.000000  
Name: class, dtype: float64
```



# Thank you

Questions?

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Project: SpaceX Falcon 9 first-stage landing prediction

GitHub repository: <https://github.com/adilsk007/IBM-Applied-Data-Science-Capstone>

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Final presentation created for IBM Applied Data Science Capstone